SAI VINEETH PALADUGU

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Electronics Engineer specializing in VLSI, Embedded Systems, and FPGA design with hands-on experience in Verilog, VHDL, C, MATLAB, and AI-driven IoT solutions. Passionate about hardware optimization, mixed-signal design, and wearable health technology.

EDUCATION

University of North Texas Aug 2024 - May 2026
M.S., Computer Engineering

National Institute of Technology Durgapur Aug 2020 - May 2024
B.Tech, Electronics and Communication Engineering

EXPERIENCE

University of North Texas | Graduate Teaching Assistant | Denton, TX Aug 2024 – May 2026

- Mentored students in digital logic design and FPGA-based system development using Xilinx Vivado and VHDL.
- Guided students through RTL design, synthesis, implementation, and timing analysis targeting Nexys 4 DDR FPGA board.
- Assisted in debugging synthesis and implementation issues, performing simulation and waveform verification.
- Delivered lab demonstrations and 1:1 sessions on FSM design, counters, and arithmetic units, ensuring clean, synthesizable VHDL coding practices.

PROJECTS

CMOS Image Sensor Design for Endoscopic Medical Applications Present

- Designed a pixel array (four-transistor architecture) in VHDL for high sensitivity in low light, improving medical imaging quality.
- Developed column amplifiers to boost weak signals while minimizing noise and improving signal integrity.
- Engineered a VHDL readout circuit for seamless interfacing between the pixel array and image-processing stage.
- Implemented a high-resolution ADC minimizing quantization error; validated functionality via mixed-signal simulations.

AI-Powered Wearable Health Monitor for Chronic Disease Management Present

- Built an embedded wearable (C/C++) tracking heart rate, SpO2, body temperature, and motion in real time.
- Integrated anomaly detection via TensorFlow Lite to enable early alerts for irregular health conditions.
- Designed a custom PCB and 3D-printed enclosure for compact, ergonomic wearability.
- Added BLE connectivity for app integration, dashboard visualization, and emergency alerts.

FIR Filter Design 2024

- Implemented high-pass, low-pass, and band-pass FIR filters in MATLAB for signal processing applications.
- Optimized coefficients using Particle Swarm Optimization (PSO) with constriction factor and inertia weight for improved frequency response.

Digital Locking System 2023

- Built a microcontroller-based locking system (Arduino UNO) with a 4x4 keypad and servo motor for password-protected access.
- Programmed authentication logic in Arduino IDE for predefined code verification.

SKILLS

Programming & HDL: Verilog, VHDL, C, MATLAB

EDA Tools: Xilinx Vivado, ModelSim, UML

Embedded Systems: Arduino, FPGA, IoT
Digital & Analog Design: CMOS circuits, ADC, FIR filters
Machine Learning: TensorFlow Lite for embedded AI